⚡ Clinoma

معسكر الورقة الثانية للأطفال — الأسئلة المقالية

✍️ Essay Booklet – Day 2 (مقالي اليوم الثاني)

✅ Answer Key Edition
Total Questions: 87 | Second Paper Camp

BOOKLET TYPE	TOTAL QUESTIONS	ACTIVE CAMP
Essay Questions (مقالي)	87 Questions	Second Paper (الورقة الثانية)

CHAPTER 01

Renal Diseases

HEMATURIA

Q1 Define hematuria and microscopic hematuria

Model Answer

- Hematuria = presence of blood in urine, either gross (visible to naked eye) or microscopic (detected only by dipstick or microscopy).
- Microscopic hematuria = more than 5 RBCs per high power field in the sediment from 10 ml of centrifuged freshly voided urine.

Q2 Enumerate the causes of hematuria in children (glomerular and extra-glomerular)

Model Answer

Glomerular:
- Isolated renal: IgA nephropathy, post-infectious GN, membranous nephropathy, membranoproliferative GN, FSGS, anti-GBM disease
- Multi-system: SLE nephritis, HSP nephritis, Goodpasture syndrome, HUS, sickle cell glomerulopathy, HIV nephropathy

Extra-glomerular:
- Upper urinary tract: pyelonephritis, interstitial nephritis, ATN, renal vein/artery thrombosis, crystalluria (calcium, oxalate, uric acid), cystic kidney disease, Wilms tumor, trauma
- Lower urinary tract: cystitis, urethritis, urolithiasis

Q3 How can you differentiate glomerular hematuria from lower urinary tract hematuria?

Model Answer

Feature	Glomerular (Renal)	Lower Urinary Tract
Color	Cola / smoky	Pink or red
RBCs morphology	Dysmorphic	Normal
RBCs casts	+	–
Blood clots	–	+
Proteinuria	>100 mg/dL	<100 mg/dL
Symptoms	Edema, hypertension	Dysuria

Q4 Enumerate the causes of dark red / brown urine

Model Answer

With RBCs (Hematuria): glomerular and extra-glomerular causes as above.
Without RBCs — Heme positive:
- Hemoglobinuria: in acute hemolytic anemia → fragmented RBCs + reticulocytosis, plasma color red
- Myoglobinuria: rhabdomyolysis (viral myositis, crush injury, prolonged fits) → high serum creatine kinase, plasma color clear
Without RBCs — Heme negative:
- Drugs: chloroquine, deferoxamine, ibuprofen, metronidazole, nitrofurantoin, rifampin, salicylates, sulfasalazine, phenazopyridine
- Dyes: beets, blackberries, food coloring
- Methemoglobin, urates

Q5 List the laboratory investigations used in the evaluation of a child with glomerular hematuria

Model Answer

For all cases:
1. Fresh urine analysis (casts, bacteria, crystals, RBCs shape)
2. Abdominal ultrasound
3. Renal function tests
4. Blood picture
5. Coagulation profile

For glomerular hematuria specifically:
- Serum C3 (reduced in: post-infectious GN, SLE nephritis, nephritis with chronic infection, MPGN)
- For post-strep GN: ASO titre, anti-DNase B titre, throat/skin culture
- For lupus: ANA, anti-double stranded DNA

Q6 Mention the indications for renal biopsy in a child with hematuria

Model Answer

1. Unexplained persistent or recurrent gross hematuria
2. Lupus nephritis
3. GN with nephritic-nephrotic picture or absent/low C3
4. Unexplained acute renal failure

📂 ACUTE POST-STREPTOCOCCAL GLOMERULONEPHRITIS (APSGN)

Q1 Define APSGN

Model Answer

Acute specific self-limited glomerulonephritis due to prior streptococcal infection, characterized by:
1. Sudden onset of gross hematuria
2. Edema (mild to moderate)
3. Hypertension
4. Oliguria (may be present)

Q2 Describe the clinical picture of APSGN

Model Answer

- Age: 5–12 years, uncommon before age 3
- Latent period: 1–2 weeks after pharyngitis, 3–6 weeks after pyoderma
- Gross hematuria (cola/smoky urine)
- Edema: results from salt and water retention
- Hypertension: seen in 60% of patients
- Oliguria may be present
- Severity ranges from asymptomatic microscopic hematuria to acute kidney injury
- Nonspecific symptoms: malaise, lethargy, abdominal/flank pain, fever
- May develop encephalopathy or heart failure due to hypertension or hypervolemia
- Acute phase resolves within 6–8 weeks
- Microscopic hematuria may persist for 1–2 years

Q3 **Enumerate the investigations used to confirm the diagnosis of APSGN**

Model Answer

Urine analysis:
- Color: smoky red / cola-like
- Oliguria with high specific gravity
- Dysmorphic RBCs
- RBC casts
- Nephritic range proteinuria
- Polymorphonuclear leukocytes

Serology:
- Serum C3: reduced in acute phase, returns to normal after 6–8 weeks
- ASO titre: elevated after pharyngeal infection
- Anti-DNase B: best marker for cutaneous streptococcal infection
- Throat or skin culture

Clinical diagnosis is likely when:
- Acute nephritic syndrome
- Evidence of recent streptococcal infection
- Low C3 level

Q4 **Enumerate the complications of APSGN**

Model Answer

- Hypertensive encephalopathy (10% of cases)
- Congestive heart failure
- Hyperkalemia
- Hyperphosphatemia
- Hypocalcemia
- Metabolic acidosis
- Seizures
- Uremia
- Rapidly progressive glomerulonephritis (RPGN)

Q5 **List the medical emergencies in APSGN**

Model Answer

A. Hyperkalemia
B. Hypertensive encephalopathy
C. Congestive heart failure
D. Rapidly progressive glomerulonephritis (RPGN)

Q6 Mention the treatment of APSGN

Model Answer

A. General measures:
1. Bed rest: only during the oliguric phase
2. Fluid restriction: intake = UOP + 400 ml/m^2/day
3. Protein and salt restriction
4. High carbohydrate diet

B. Specific measures:
1. Management of renal insufficiency and hypertension
2. Eradication of streptococcal infection: 10-day course of penicillin (does not alter natural history of GN)

📂 NEPHROTIC SYNDROME

Q1 Define nephrotic syndrome and mention its four characteristic features

Model Answer

Nephrotic syndrome is primarily a pediatric disorder, 15 times more common in children than adults (incidence 2–3/100,000 children/year).
Four characteristic features:
1. Heavy proteinuria (>3.5 g/24hr in adults or >40 mg/m²/hr in children)
2. Hypoalbuminemia (<2.5 g/dL)
3. Generalized edema
4. Hyperlipidemia (hypercholesterolemia >250 mg/dL)

Q2 Describe the pathogenesis of edema in nephrotic syndrome

Model Answer

1. Increased GBM permeability → heavy proteinuria → hypoalbuminemia
2. Decreased plasma oncotic pressure → fluid shift to interstitial tissue
3. Decreased intravascular volume → decreased renal blood flow → activation of JGA → increased aldosterone → Na and H_2O retention → more edema
4. Decreased intravascular volume → increased ADH → H_2O retention → more edema
5. Hypoproteinemia → liver increases lipoprotein synthesis → hyperlipidemia (+ decreased lipoprotein lipase → decreased lipid catabolism)

Q3 Enumerate the complications of nephrotic syndrome

Model Answer

1. Infection (major complication): spontaneous bacterial peritonitis most common (S. pneumoniae commonest organism), also sepsis, pneumonia, cellulitis, UTI
2. Thromboembolic events (2–5% in children): renal vein thrombosis, pulmonary embolus, sagittal sinus thrombosis
3. Hypovolemic shock (with aggressive diuretic therapy)
4. Renal failure (from hypovolemia or renal vein thrombosis)
5. Relapse
6. Complications of therapy (steroids, cyclophosphamide)

Q4 Describe the steroid therapy used in idiopathic nephrotic syndrome

Model Answer

Induction of remission:
- Prednisone 60 mg/m^2/day (max 60 mg/day), divided into 2–3 doses, for at least 4 consecutive weeks
- 80–90% of children respond (urine trace/negative for protein for 3 consecutive days = steroid responsive)
Alternate-day dose:
- After initial 4–6 weeks: taper to 40 mg/m^2/day given every other day as single morning dose, then slowly discontinued over 2–3 months
Definitions:
- Steroid dependent: relapses while on alternate-day therapy OR within 14 days of stopping prednisone
- Frequent relapsers: respond well but relapse 4 or more times in 12 months
- Steroid resistant: fail to respond within 4–6 weeks → renal biopsy required
Alternative agents (for steroid-dependent/resistant):
Cyclophosphamide, methylprednisolone, cyclosporine, ACE inhibitors + angiotensin II blockers

Q5 List the indications for renal biopsy in nephrotic syndrome

Model Answer

Before treatment:
1. Hematuria
2. Hypertension
3. Renal insufficiency
4. Hypocomplementemia (low C3)
5. Age <1 year or >10 years
After treatment:
1. Frequent relapsers
2. Steroid resistant

Q6 Mention the indications for IV albumin infusion in nephrotic syndrome

Model Answer

IV 25% human albumin (0.5 g/kg/12hr):
1. Massive generalized edema (anasarca) with respiratory difficulty → albumin + IV furosemide (1–2 mg/kg)
2. Hypovolemic shock → albumin alone
(Close monitoring of volume status, serum electrolytes, and renal function is mandatory)

📂 ACUTE KIDNEY INJURY (AKI)

Q1 Define AKI, oliguria, and anuria in children

Model Answer

- **AKI:** rapid decline (hours to days) in GFR resulting in: impairment of nitrogenous waste excretion, loss of water/electrolyte regulation, and loss of acid-base regulation
- **Oliguria:** urine output <1 ml/kg/hr or <400 ml/m^2/day
- **Anuria:** urine output <30 ml/m^2/day

Q2 Enumerate the causes of AKI classified into prerenal, intrinsic renal, and postrenal

Model Answer

Prerenal:
Dehydration, hemorrhage, sepsis, hypoalbuminemia, cardiac failure
Intrinsic renal:
- Glomerulonephritis: post-infectious GN, SLE, HSP, MPGN, anti-GBM disease, HUS
- Acute tubular necrosis, cortical necrosis, renal vein thrombosis
- Acute interstitial nephritis, tumor infiltration, tumor lysis syndrome
Postrenal:
Posterior urethral valve, PUJ obstruction, ureterovesical junction obstruction, ureterocele, tumors, urolithiasis, hemorrhagic cystitis, neurogenic bladder

Q3 How do you differentiate prerenal AKI from intrinsic renal AKI?

Model Answer

Feature	Prerenal	Intrinsic Renal
History	Vomiting, diarrhea, dehydration	Pharyngitis + edema + hematuria; or hypotension + nephrotoxic drugs
Examination	Tachycardia, dry mucosae, poor perfusion	Peripheral edema, rales, gallop; ± rash/arthritis
Specific gravity	>1020	<1010
Urine osmolality	>500 mOsm/kg	<350 mOsm/kg
Urine Na	<20 mEq/L	>40 mEq/L
FENa	<1% (<2.5% neonates)	>2% (>10% neonates)

Q4 Enumerate the clinical manifestations of AKI (oliguric phase)

Model Answer

Early AKI:
- Oliguria or anuria
- Edema
- Hypertension
- Acidotic breathing (rapid + deep)

Advanced AKI (above +):
- ↑ Urea → uremic encephalopathy (confusion → convulsion → coma)
- ↓ Na → convulsions
- ↑ K → arrhythmias
- Hypervolemia → heart failure → acute pulmonary edema
- Metabolic acidosis → GIT stress ulcer

Q5 List the stages of AKI according to modified pRIFLE criteria

Model Answer

Stage	GFR	Urine Output
Risk	GFR decrease by 25%	<0.5 ml/kg/hr for 8 hrs
Injury	GFR decrease by 50%	<0.5 ml/kg/hr for 16 hrs
Failure	GFR decrease by 75% or <35 ml/min/1.73m^2	<0.3 ml/kg/hr for 24 hrs or anuric for 12 hrs
Loss	Persistent failure >4 weeks	—
End stage	Persistent failure >3 months (ESRD)	—

Q6 Mention the electrolyte and acid-base disturbances seen in advanced AKI

Model Answer

- Hyperkalemia → arrhythmias
- Dilutional hyponatremia → convulsions
- Hypocalcemia
- Hyperphosphatemia
- Metabolic acidosis (↓ pH, ↓ $PaCO_2$, ↓ HCO_3)
- ↑ Urea → uremic encephalopathy

CHAPTER 02

Neonatology

📂 Topic 1: Transient Cutaneous Lesions

Q1 Define transient cutaneous lesions in the newborn

Model Answer

Transient cutaneous lesions are temporary, self-limiting skin changes seen in the normal newborn due to physiological adaptation to extrauterine life. They require no treatment and resolve spontaneously.

Q2 Enumerate the transient cutaneous lesions seen in the newborn

Model Answer

* Acrocyanosis
* Mottling (Cutis Marmorata)
* Harlequin color change
* Vernix caseosa
* Lanugo hair
* Salmon patch (capillary hemangioma)
* Mongolian spots
* Erythema toxicum
* Pustular melanosis
* Milia of face
* Physiological jaundice

Q3 Define Acrocyanosis

Model Answer

Temporary, harmless cyanosis of the hands and feet seen in newborns, especially when they are cool, resulting from vasomotor instability and peripheral circulatory sluggishness.

Q4 Define Harlequin color change

Model Answer

A rare, transient, and harmless division of the newborn's body from the forehead to the pubis into red and pale halves, which characteristically occurs when the neonate lies on his side.

Q5 Define Erythema toxicum

Model Answer

Small, white, occasionally vesiculopustular papules on an erythematous base that develop 1–3 days after birth, persist for as long as 1 week, contain eosinophils, and are usually distributed on the face, trunk, and extremities.

Q6 Mention the clinical features of Erythema toxicum

Model Answer

* Appears 1–3 days after birth.
* Small white vesiculopustular papules on an erythematous base.
* Contains eosinophils (not neutrophils).
* Distributed on face, trunk, and extremities.
* Persists up to 1 week then disappears spontaneously.

Q7 Define Mongolian spots

Model Answer

Blue, well-demarcated areas of pigmentation seen over the buttocks, back, and sometimes other parts of the body in more than 50% of newborns, which tend to disappear within the first year.

Q8 Mention the clinical features of Mongolian spots

Model Answer

* Blue, well-demarcated area of pigmentation.
* Seen over buttocks, back, and sometimes other parts of the body.
* Seen in more than 50% of newborns.
* Tend to disappear within the first year of life.
* Treatment: reassurance only.

Q9 Mention the clinical features of Pustular Melanosis

Model Answer

* Benign lesion seen predominantly in black neonates.
* Present at birth as a vesiculopustular eruption.
* Distributed around the chin, neck, back, extremities, palms, or soles.
* Contains neutrophils (unlike erythema toxicum).
* Lasts for 2–3 days.

Q10 Differentiate between Erythema toxicum and Pustular Melanosis

Model Answer

* **Erythema toxicum:** Appears 1–3 days after birth, contains eosinophils, affects face/trunk/extremities, and resolves in 1 week.
* **Pustular melanosis:** Present at birth, contains neutrophils, seen predominantly in black neonates, affects chin/neck/back/extremities/palms/soles, and lasts 2–3 days.

Q11 List four transient skin features or substances related to maturity or delivery found on a newborn's skin

Model Answer

* Vernix caseosa (cheesy material, decreases near term).
* Lanugo hair (fine immature hair, lost or replaced in term infants).
* Scattered petechiae (seen on the presenting part after difficult delivery).
* Amniotic bands (associated with membrane rupture or vascular compromise).

📂 Topic 2: Prematurity and Its Complications

Q1 Define Prematurity / Premature infant

Model Answer

Prematurity refers to a live-born infant delivered before 37 completed weeks of gestational age, as defined by the World Health Organization.

Q2 Enumerate the causes of prematurity

Model Answer

* **Idiopathic:** Most common cause.
* **Maternal causes:** Young age (<20 yrs) or older age (>40 yrs), pre-eclampsia, abruptio placentae, placenta praevia, chorioamnionitis, PROM, polyhydramnios, incompetent OS, trauma, DM, chronic illness, and drug abuse.
* **Fetal causes:** Fetal distress, fetal anomalies, multiple gestation, erythroblastosis, and non-immune hydrops.
* **Socioeconomic:** Low socioeconomic status.

Q3 Enumerate the physiological handicaps of prematurity

Model Answer

* **Impaired thermoregulation:** Due to immature centers, large surface area, and reduced subcutaneous fat.
* **Respiratory:** Weak muscles, surfactant deficiency, and poor cough/gag reflexes.
* **GIT:** Poor sucking/swallowing, weak cardiac sphincter, and decreased fat absorption.
* **Hepatic:** Impaired bilirubin conjugation and vitamin K-dependent factor deficiency.
* **Haemostatic:** Increased capillary fragility and impaired clotting mechanisms.
* **Renal:** Low GFR and poor urine concentration capability.
* **CNS:** Poorly developed brain, soft skull, and immature blood-brain barrier.
* **Immunological:** Reduced placental transmission of immunoglobulins and impaired phagocytosis.

Q4 **Enumerate the physiological handicaps of the premature infant specifically regarding Respiratory and Gastrointestinal systems**

Model Answer

* **Respiratory Handicaps:** Poorly developed respiratory center, weak respiratory muscles, deficiency of pulmonary surfactant, poorly developed lung alveoli, and poor gag and cough reflexes.
* **Gastrointestinal Handicaps:** Poor suckling and swallowing reflexes, weak cardiac sphincter, low stomach capacity and acidity, decreased fat digestion/absorption of fats and fat-soluble vitamins, and hypotonicity of the muscle coat of the colon.

Q5 **List four physical features or signs of underdevelopment observed during the clinical examination of a premature infant**

Model Answer

* Underdeveloped ear cartilage (soft ear firmness).
* Absent or minimal sole creases (creases on anterior 1/3 or before).
* Immature genitalia (testis in inguinal canal / prominent labia minora).
* Diminished or absent breast nodule (breast size < 0.5 cm).

Q6 **List the Respiratory complications of prematurity**

Model Answer

* Respiratory distress syndrome (hyaline membrane disease)
* Bronchopulmonary dysplasia
* Pneumothorax and pneumomediastinum
* Pulmonary interstitial emphysema
* Congenital pneumonia
* Pulmonary hypoplasia
* Pulmonary hemorrhage
* Apnea of prematurity

Q7 **List the Central Nervous System (CNS) complications of prematurity**

Model Answer

* Intraventricular hemorrhage (IVH)
* Periventricular leukomalacia
* Hypoxic-ischemic encephalopathy (HIE)
* Seizures
* Retinopathy of prematurity (ROP)
* Deafness
* Kernicterus (bilirubin encephalopathy)

Q8 Enumerate the Cardiovascular and Gastrointestinal complications of prematurity

Model Answer

* **Cardiovascular Complications:** Patent ductus arteriosus (PDA), hypotension, hypertension, and bradycardia (associated with apnea).
* **Gastrointestinal Complications:** Difficult feeding, necrotizing enterocolitis (NEC), hyperbilirubinemia (direct and indirect), and spontaneous isolated gastrointestinal perforation.

Q9 Mention the Hematologic complications of prematurity

Model Answer

* Anemia (early or late onset)
* Hyperbilirubinemia-indirect
* Subcutaneous or organ (liver, adrenal) hemorrhage
* Vitamin K deficiency / Disseminated intravascular coagulopathy (DIC)

Q10 List four Metabolic and Endocrine complications of prematurity

Model Answer

* Hypocalcemia
* Hypoglycemia
* Hypothermia
* Electrolyte disturbances

Q11 Mention the essential guidelines for prophylactic vitamin administration in the management of a preterm infant

Model Answer

* **Vitamin K1:** 1 mg IM once immediately after birth.
* **Vitamin D:** 800 I.U./day.
* **Vitamin A:** 8000 I.U./day (usually started after the 2nd week).
* **Vitamin C:** 50 mg/day.
* **Folic Acid:** 5 mg twice weekly.

📂 Topic 3: Compare Between Physiological and Pathological Jaundice

Q1 Define the Enterohepatic Circulation of Bilirubin

Model Answer

The process where conjugated bilirubin in the small intestine is converted back to the unconjugated form by the action of the enzyme β-glucuronidase, and is subsequently reabsorbed into the circulation to undergo reconjugation in the liver.

Q2 Define and state the criteria of Physiological Jaundice

Model Answer

Physiological jaundice is a self-limiting, benign hyperbilirubinemia of the newborn.
* **In full-term infants:** It appears on the 2nd–3rd day of life, peak bilirubin level is <12 mg/dL, and it resolves within 7 days.
* **In preterm infants:** It appears on the 4th–6th day, peak level is <14 mg/dL, and resolves within 10–14 days.
* **Clinical status:** The infant remains completely傲 asymptomatic and requires no treatment.

Q3 List the predisposing factors/causes that explain the development of physiological jaundice

Model Answer

* Increased bilirubin production from the accelerated breakdown of fetal red blood cells.
* Transient limitation in the hepatic uptake of bilirubin (deficiency of Y and Z ligandin proteins).
* Transient limitation in bilirubin conjugation due to immature glucuronyl transferase (G.T.) enzyme.
* Increased enterohepatic circulation of bilirubin.

Q4 Enumerate the risk factors for exaggerated physiological jaundice

Model Answer

* Maternal diabetes.
* Prematurity.
* Race (Chinese, Japanese, Korean, and Native American).
* Male sex.
* Polycythemia or Trisomy 21.
* Cephalohematoma or cutaneous bruising.
* Oxytocin induction.
* Weight loss (dehydration or caloric deprivation) and delayed bowel movement.
* A sibling who had physiological jaundice.

Q5 Define Pathological Jaundice and list the criteria used to identify it

Model Answer

Pathological jaundice is neonatal jaundice that does not fit the typical criteria for physiological jaundice. It is highly suspected in the presence of any of the following criteria:
* Appears within the first 24 hours of life.
* Serum bilirubin is rising at a rate faster than 5 mg/dL/24 hr.
* Serum bilirubin exceeds 12 mg/dL in full-term or 14 mg/dL in preterm infants.
* Jaundice persists beyond 7 days in full-term or 14 days in preterm infants.
* Direct-reacting bilirubin is greater than 20% of the total bilirubin at any time.
* Family history of hemolytic disease, or presence of pallor, hepatomegaly, and splenomegaly.
* Failure of phototherapy to lower bilirubin, or the infant shows signs of illness or kernicterus.

Q6 Mention four clinical signs of an "Ill baby" with Pathologic Jaundice

Model Answer

* Vomiting or poor feeding.
* Lethargy or apnea.
* Hepatomegaly or splenomegaly.
* Light-colored stools or dark urine positive for bilirubin.

Q7 Enumerate four Pathologic causes of Neonatal Jaundice due to "Increased Formation" (Hemolysis), excluding immune incompatibility

Model Answer

* **RBC cell wall defects:** Congenital spherocytosis or elliptocytosis.
* **Enzymatic defects:** Glucose-6-phosphate dehydrogenase (G6PD) deficiency or pyruvate kinase deficiency.
* **Hemoglobin defects:** α-thalassemia.
* **Extravascular hemolysis:** Cephalhematoma or internal tissue bruising.

Q8 Enumerate four causes of Pathologic Jaundice due to Conjugation Defects

Model Answer

* Immaturity of G.T. enzyme (e.g., Hypothyroidism).
* Crigler-Najjar Syndrome Type I (severe deficiency).
* Crigler-Najjar Syndrome Type II (mild/moderate deficiency) or Gilbert's Syndrome.
* Inhibition of the enzyme (e.g., Breast milk jaundice or Lucey Driscoll syndrome).

Q9 Mention four causes of Pathologic Jaundice related to Excretion or Secretion Defects

Model Answer

* Biliary Atresia (obliteration of the intrahepatic or extrahepatic biliary tree).
* Choledochal Cyst.
* Dubin-Johnson Syndrome or Rotor Syndrome (hepatocyte secretion defects).
* Inspissated Bile Syndrome.

📂 Topic 4: Complications of Indirect Hyperbilirubinemia

Q1 Enumerate the complications of indirect hyperbilirubinemia

Model Answer

* Inspissated bile syndrome.
* Kernicterus (bilirubin encephalopathy).

Q2 Define Inspissated Bile Syndrome

Model Answer

A complication of indirect hyperbilirubinemia characterized by persistent icterus in association with significant elevations in both direct and indirect bilirubin levels in infants with hemolytic disease, which clears spontaneously within a few weeks or months.

Q3 Define Kernicterus

Model Answer

Kernicterus, or bilirubin encephalopathy, is a neurological syndrome resulting from the pathological deposition of unconjugated bilirubin in the basal ganglia and brainstem nuclei. It typically develops when serum bilirubin levels exceed 30 mg/dL (overall danger range 21–50 mg/dL).

Q4 Enumerate the predisposing factors for kernicterus

Model Answer

* Unconjugated hyperbilirubinemia (markedly elevated indirect bilirubin).
* Hypoalbuminaemia (which increases the toxic free form of unbound bilirubin).
* Factors that increase the permeability of the Blood-Brain Barrier (B.B.B), such as sepsis, hypoxia, and acidosis.

Q5 **Describe the clinical stages of Acute Kernicterus**

Model Answer

* **Phase 1 (1st 1–2 days):** Poor sucking, loss of Moro reflex, stupor, high-pitched cry, hypotonia, and seizures.
* **Phase 2 (middle of 1st week):** Hypertonia of extensor muscles, opisthotonos, retrocollis, bulging fontanel, and fever.
* **Phase 3 (after 1st week):** Manifests primarily as persistent hypertonia.

Q6 **List the clinical features of Chronic Kernicterus (by 3 years of age) and during the 2nd year of life**

Model Answer

* **During the 2nd year of life:** Abatement of opisthotonos and seizures, accompanied by a steady increase in irregular involuntary movements, muscle rigidity, or hypotonia.
* **By 3 years of age (Complete Syndrome):**
* Bilateral choreoathetosis and extrapyramidal signs.
* Seizures and mental deficiency.
* Dysarthric speech and high-frequency hearing loss.
* Squinting and defective upward movement of the eyes.
* Pyramidal signs, hypotonia, and ataxia (in some infants).

Q7 **Mention four preventable causes/pitfalls leading to Kernicterus**

Model Answer

* Early discharge (<48 hr) with no early follow-up (within 48 hr of discharge), particularly in near-term infants.
* Failure to check the bilirubin level in an infant noted to be jaundiced in the first 24 hours.
* Underestimating the severity of jaundice by clinical (visual) assessment alone.
* Delay in measuring the serum bilirubin level or delay in initiating phototherapy/exchange transfusion.

Q8 **List the parameters to monitor and evaluate when performing a visual/clinical assessment of a jaundiced newborn**

Model Answer

* **Anatomic progression of jaundice via dermal pressure:** Progresses from face ($\approx$ 5 mg/dL) to mid-abdomen ($\approx$ 15 mg/dL) and down to the soles ($\approx$ 20 mg/dL).
* **Type/hue of jaundice:** Bright yellow or orange color signifies indirect bilirubin deposition, whereas a greenish color indicates obstructive (direct) bilirubin.
* **Vascular/tissue signs:** Presence of pallor, plethora, or extravascular blood (cephalohematoma, bruising).
* **Organ systemic signs:** Hepatosplenomegaly (H.S.M.), signs of congenital infection, abdominal distention, or central nervous system affection.

Q9 **Enumerate four major complications or risks associated with performing an Exchange Transfusion**

Model Answer

* Transient bradycardia (with or without calcium infusion) or cyanosis/apnea.
* Electrolyte disturbances and metabolic acidosis.
* Vascular thrombosis or portal vein thrombosis leading to late portal hypertension.
* Necrotizing enterocolitis (NEC).

📂 Topic 5: Neonatal Sepsis and Its Clinical Sepsis Score

Q1 Define Neonatal Sepsis

Model Answer

Neonatal sepsis is a clinical syndrome of systemic illness resulting from the pathophysiological effects of local or systemic infection accompanied by bacteremia in the first 4 weeks of life. In 30% of cases it is associated with meningitis, and in 20% of cases with pneumonia or urinary tract infection.

Q2 Define Nosocomial Sepsis

Model Answer

Hospital-acquired neonatal sepsis where clinical manifestations appear after about 8 days from hospital admission, with a pathogenesis directly related to the underlying neonatal illness and the unique flora of the neonatal intensive care unit (NICU).

Q3 Differentiate between Early onset and Late onset neonatal sepsis

Model Answer

* **Early onset sepsis:** Manifestations are present at birth or anytime during the first week (usually <3 days) of life; it is a fulminant multisystem illness where the organism is acquired during the intrapartum period from the maternal genital tract (or transplacentally). History of maternal obstetric complications is common.
* **Late onset sepsis:** Manifestations appear 8–28 days after delivery; it can be multisystem or focal. The organism is acquired from the maternal genital tract or the environment (human contact/contaminated equipment). Maternal obstetric complications are uncommon.

Q4 Differentiate between Possible, Probable, and Proven Sepsis based on diagnostic classification

Model Answer

* **Possible Sepsis:** Defined by the isolated presence of suggestive clinical signs and symptoms.
* **Probable Sepsis:** Suggestive clinical signs coupled with risk factors, a positive septic screen (CBC, CRP, ESR), or radiological evidence of pneumonia.
* **Proven Sepsis:** Clinical signs of sepsis confirmed by a positive culture from a normally sterile site (blood, CSF, or urine).

Q5 **Enumerate the risk factors for neonatal sepsis**

Model Answer

* **Maternal risk factors:** Premature rupture of membranes (>24 hr), maternal peripartum fever ($\ge$ 38.0°C) or chorioamnionitis, UTI, vaginal/perineal colonization with GBS/E.coli, and meconium-stained or foul-smelling amniotic fluid.
* **Neonatal risk factors:** Prematurity (single most important risk factor), multiple gestations, galactosemia (predisposed to E.coli), immune defects, resuscitation at birth, male sex, and black race.
* **Environmental risk factors:** Crowding, inadequate nursing staff, lack of hand washing, invasive procedures, and prolonged antibiotic use.

Q6 **List the causative organisms of neonatal sepsis**

Model Answer

* E. coli
* Klebsiella
* Enterococci
* Proteus
* Pseudomonas
* Listeria monocytogenes
* Group B Streptococcus (GBS)

Q7 **Mention the Clinical Sepsis Score for neonatal sepsis**

Model Answer

Sepsis is suspected if a baby has 3 or more of the following signs (each item scores 1 point):
* Apnea, tachypnea, cyanosis, respiratory distress — 1 point.
* Bradycardia or tachycardia — 1 point.
* Hypotonia or seizure — 1 point.
* Poor skin color, poor peripheral circulation — 1 point.
* Irritability, lethargy, poor feeding — 1 point.
* Hepatosplenomegaly, abdominal distension — 1 point.
* Fever or hypothermia — 1 point.

Q8 List four clinical signs of Central Nervous System (CNS) and Gastrointestinal (GI) dysfunction in a septic neonate

Model Answer

* **CNS Signs:** Lethargy, irritability, tremors, seizures, hypotonia, hyporeflexia, a full fontanelle, or a high-pitched cry.
* **Gastrointestinal Signs:** Abdominal distention, vomiting, poor feeding, diarrhea, hepatomegaly, or splenomegaly.

Q9 Enumerate the investigations (sepsis work-up) for neonatal sepsis

Model Answer

* **Evidence of infection:** Blood culture (gold standard), CSF culture, urine culture (by catheter or suprapubic puncture for neonates > 24h), and Gram stain of amniotic fluid or gastric aspirate.
* **Evidence of inflammation:** * CBC (neutrophilia/neutropenia, band cell to total neutrophil ratio >0.2, toxic granulation, thrombocytopenia, fragmented RBCs).
* Elevated acute-phase reactants (CRP and ESR) and cytokines (Interleukin-6).
* **Evidence of Multiorgan Dysfunction:** Blood gases (metabolic acidosis), renal function tests (BUN, creatinine), hepatic function tests (bilirubin, ALT, AST, ammonia, PT, PTT for DIC).
* **Radiological studies:** Chest X-ray (for respiratory symptoms) and urinary tract imaging (renal ultrasound/scan).

CHAPTER 03

Family Medicine

📂 Title 7 & 8: Anticipatory care (1 & 2)

Q1 **Enumerate the three distinct levels of disease prevention included under Anticipatory Care, along with the core definition or objective of each level**

📖 **Case Scenario:**
Case 7:
A family physician is conducting a periodic health examination for an adolescent patient. The physician aims to implement systematic, evidence-based measures to promote good health, screen for underlying psychosocial risk factors using the HEEADSSS framework, and carefully verify vaccination histories and clinical baselines.
*

Model Answer

1. **Primary Prevention:** Prevention of disease development. This includes providing health education about conditions known to cause disease, and prophylaxis/vaccination.
2. **Secondary Prevention:** Early diagnosis and treatment. This involves screening to detect undeclared disease in外 asymptomatic, apparently healthy populations, and case finding.
3. **Tertiary Prevention:** Systematic, long-term monitoring of a patient to prevent or minimize the impact of disease complications.
*

Q2 **Enumerate the six essential operational elements that a family physician must comprehensively evaluate to build a proper database during a periodic health examination**

Model Answer

1. Building the database.
2. Medical history.
3. Family history.
4. Social history.
5. Occupational and environmental history.
6. Anticipatory guidance.
*

Q3 **Enumerate the specific clinical parameters that correspond to each letter of the **HEEADSSS** psychosocial assessment interview framework used during adolescent screening**

Model Answer

1. **H** - Home.
2. **E** - Eating.
3. **E** - Education.
4. **A** - Activities.
5. **D** - Drugs.
6. **S** - Suicide/depression.
7. **S** - Sex.
8. **S** - Safety.
*

Q4 **List four true clinical contraindications to implementing childhood immunizations in a family practice setting**

Model Answer

1. Anaphylactic reaction to a vaccine.
2. Seizure or fever $> 40.5^\circ\text{C}$ within 48 hours of a pertussis vaccine.
3. True Egg Allergy or Neomycin allergy (for MMR vaccine).
4. Immunocompromised status of the patient (for OPV vaccine) OR untreated moderate to severe illness associated with fever.

📂 Title 9: Breast feeding

Q5 Enumerate the four clinical scenarios where a mother must **ABSOLUTELY NOT** breastfeed and **NOT** feed expressed breast milk to her infant

📖 **Case Scenario:**
Case 8:
A mother presents to the family medicine center for a postpartum consultation. The family physician systematically assesses her medical history to ensure there are no infectious, metabolic, or pharmacological conditions that would serve as an absolute or temporary contraindication to breastfeeding or feeding expressed breast milk.
*

Model Answer

1. Infant is diagnosed with classic galactosemia (a rare genetic metabolic disorder).
2. Mother has HIV and is not on antiretroviral therapy (ART).
3. Mother is using an illicit drug, such as opioids, PCP (phencyclidine), or cocaine.
4. Mother has suspected or confirmed Ebola virus disease.
*

Q6 Enumerate three specific conditions under which a mother must **TEMPORARILY NOT** breastfeed and **NOT** feed expressed breast milk to her infant, where she must wait until a physician determines her milk is safe

Model Answer

1. Mother is infected with untreated brucellosis.
2. Mother is undergoing diagnostic imaging with radiopharmaceuticals.
3. Mother has an active herpes simplex virus (HSV) infection with lesions present on the breast.
*

Q7 State the specific clinical scenario where a mother must **TEMPORARILY NOT** breastfeed but **CAN** still feed expressed breast milk to her infant, including the required precautions and the timeline for safely resuming direct breastfeeding

Model Answer

* **The Condition:** Mother has untreated, active tuberculosis.
*
Required Precautions: Airborne and contact precautions that may require temporary separation of the mother and infant, during which expressed breast milk should be given.
*
Timeline for Resuming Breastfeeding: The mother may resume breastfeeding once she has been treated appropriately for 2 weeks and is documented to be no longer contagious.

📁 Topic A: Possible Bacterial Infection

Q8 **Enumerate the clinical signs that classify a young infant as having a **"LOCAL BACTERIAL INFECTION"** under the IMCI protocols**

📖 **Case Scenario:**
Case 9:
A 3-week-old young infant is brought to the primary care unit. The family physician utilizes the structured IMCI guidelines to screen the infant for clinical signs of a localized or systemic bacterial infection.
*

Model Answer

(Presence of any of the following signs):
1. Pus draining from the ear.
2. Pus draining from the eyes associated with redness and swelling.
3. Umbilical redness extending to the skin OR red umbilicus or draining pus.
4. Skin pustules.
*

Q9 **Enumerate six critical clinical signs that categorize a young infant under **"POSSIBLE SERIOUS BACTERIAL INFECTION"**, necessitating urgent pre-referral stabilization and hospital referral**

Model Answer

(Presence of any of the following signs):
1. Not able to feed OR shows less than normal clinical movements.
2. Convulsions (presenting now or by history).
3. Fast breathing (≥ 60 breaths per minute).
4. Severe chest indrawing, nasal flaring, grunting, or wheeze.
5. Bulging fontanelle.
6. High fever ($\ge 37.5^\circ\text{C}$ or feels hot) OR low body temperature (less than 35.5°C or feels cold).

📂 Topic B: Significant Jaundice

Q10 **Enumerate the specific clinical signs and thresholds that classify a young infant under **"SIGNIFICANT JAUNDICE"** according to IMCI guidelines**

📖 **Case Scenario:**
Case 10:
A mother brings her 10-day-old young infant to the family medicine center because she notes progressive yellowing of the skin. The physician checks the infant's age, sclera, and limbs to determine if the clinical presentation constitutes significant jaundice.
*

Model Answer

(Presence of any of the following parameters):
1. Jaundice started in the first 24 hours of life.
2. Deep jaundice seen clearly in the sclera.
3. Jaundice extending directly to the palms and/or soles.
4. The infant's chronological age is 2 weeks or more.

📂 Topic C: Pneumonia (Cough or Difficult Breathing)

Q11 **Define the specific age-based respiratory thresholds used to classify **"Fast Breathing"** under IMCI guidelines for a child aged 2 months up to 11 months, and for a child aged 12 months up to 5 years**

📖 **Case Scenario:**
Case 11:
A 14-month-old child weighing 11 kg is brought to the clinic presenting with an acute cough. The child is calm, and the physician counts the respiratory rate over one full minute and evaluates the chest wall for any retractions.
*

Model Answer

1. **For an infant aged 2 months up to 11 months:** Fast breathing is defined as **50 breaths per minute or more**.
2. **For a child aged 12 months up to 5 years:** Fast breathing is defined as **40 breaths per minute or more**.
*

Q12 **Enumerate the clinical signs that explicitly classify a coughing child under **"SEVERE PNEUMONIA OR VERY SEVERE DISEASE"**, requiring urgent pre-referral antibiotic delivery and hospital referral**

Model Answer

(Presence of any of the following features):
1. Any general danger sign (unable to drink/breastfeed, vomits everything, history of convulsions, lethargic or unconscious).
2. Stridor present in a calm child.
3. Chest indrawing.

📂 Topic D: Diarrhea

Q13 Enumerate the specific clinical signs used to classify a child under **"SOME DEHYDRATION"** according to the IMCI diarrhea assessment chart

📖 **Case Scenario:**
Case 12:
A 2-year-old toddler presents with a 3-day history of frequent loose, watery stools. The child is restless and highly irritable, and the family physician performs an abdominal skin pinch test to determine the correct rehydration protocol.
*

Model Answer

(Requires at least two of the following clinical signs):
1. Restless, irritable.
2. Sunken eyes.
3. Drinks eagerly, thirsty.
4. Skin pinch goes back slowly.
*

Q14 Enumerate the clinical parameters that define **"SEVERE DEHYDRATION"**, and outline the four operational flowchart steps to manage this condition if intravenous (IV) fluid access is completely unavailable nearby

Model Answer

* **Clinical signs of Severe Dehydration (Requires two of the following):** Lethargic or unconscious; sunken eyes; not able to drink or drinking poorly; skin pinch goes back *very slowly* (longer than 2 seconds).
* **Four operational flowchart steps if IV fluid is NOT available nearby (Plan C flowchart):**
1. Check if you are trained to use a naso-gastric (NG) tube for rehydration.
2. If trained, start rehydration via the NG tube (or mouth) with ORS solution, giving 20 ml/kg/hour for 6 hours (total of 120 ml/kg).
3. If not trained in NG tube rehydration, immediately check if the child can drink.
4. If the child cannot drink, immediately **refer URGENTLY to the hospital** for IV or NG treatment.